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Title Microbial Bio-inoculation Effects on Seed Germination Dynamics and Field Performance of Pea (Pisum sativum L.) under Osmotic Stress and Fertilization

Authors Francisco Guevara-Fernández * , Sebastian Casas-Niño , Milagros Ninoska Munoz-Salas , Wagner Meza-Maicelo , Manuel Oliva-Cruz , Flavio Lozano-Isla *

Section Sustainable Bioresource and Bioprocess Engineering

Abstract Microbial bio-inoculants have been proposed as management tools to enhance crop performance under variable environmental conditions; however, their effectiveness is frequently modulated by site-specific factors. This study evaluated the effects of bio-inoculation on seed germination and seedling vigor of pea under osmotic stress induced by polyethylene glycol (PEG 6000), and assessed the interaction between bio-inoculation and two fertilization levels under field conditions in the Amazonas region of Peru. Under laboratory conditions, germination percentage remained high across all treatments (93.3–100%) and was not affected by bio-inoculation or osmotic potential. In contrast, osmotic stress significantly altered germination dynamics. Mean germination time increased from 1.85–2.09 days at 0 MPa to 2.26–2.43 days at −0.8 MPa, while germination synchrony and seedling vigor declined with increasing stress. The seedling vigor index reached maximum values at −0.2 MPa (4.47–5.29) and decreased at −0.8 MPa (1.50–2.00). Multivariate analyses indicated that variation in germination responses was mainly associated with germination timing and vigor rather than seed viability. Field experiments revealed variability in agronomic traits between locations. Plant height ranged from 38.5–46.3 cm in Lamud and from 100.6–108.3 cm in Molinopampa, while grain yield varied from 698–1846 kg/ha and 8771–9919 kg/ha, respectively. Yield variation was explained by pod weight and number of pods per plant. Environmental conditions showed a stronger influence than microbial bio-inoculation on germination dynamics and field productivity, although positive trends associated with bio-inoculant application were observed.

Review Report Form

Quality of English Language
( ) The English could be improved to more clearly express the research.
(x) The English is fine and does not require any improvement.

	Yes	Can be improved	Must be improved	Not applicable
Does the introduction provide sufficient background and include all relevant references?	( )	(x)	( )	( )
Is the research design appropriate?	( )	(x)	( )	( )

Are the methods adequately described?	( )	(x)	( )	( )
Are the results clearly presented?	( )	(x)	( )	( )
Are the conclusions supported by the results?	( )	(x)	( )	( )
Are all figures and tables clear and well-presented?	( )	(x)	( )	( )

Comments and Suggestions for Authors

Dear authors, it is a pleasure to receive your manuscript entitled "Microbial Bio-inoculation Effects on Seed Germination Dynamics and Field Performance of Pea (Pisum sativum L.) under Osmotic Stress and Fertilization." It is a very well-structured and highly relevant study regarding the potential of microbial bio-inoculation to improve the agronomic performance of pea (*Pisum sativum* L.), from germination to field yield. The study combines laboratory assays under simulated osmotic stress using PEG-6000 with field experiments conducted at two locations with contrasting conditions in the Peruvian Amazon, evaluating growth, phenology, and yield under different fertilization levels. This integration makes it possible to analyze crop responses both under controlled conditions and within real productive environments. As with any excellent article, some aspects should be adjusted to further improve its quality. From my perspective, the following comments will enhance the relevance and clarity of the work:

It is recommended to strengthen the regional agronomic justification of the study by explaining more clearly the importance of the crop and the use of bio-inoculants in the production systems of the region where the research was conducted (Lines 44–47). Likewise, it would be advisable to incorporate references from recent meta-analyses or reviews on bio-inoculation in legumes in order to better contextualize the work within the current state of scientific knowledge and reinforce the relevance of the study (Lines 79–85).

The germination experiment was conducted with 30 seeds per treatment, which could be limited for detecting small effects or subtle variations among treatments in factorial germination experiments. It is necessary to justify the sample size used or briefly discuss how this number of experimental units could influence the statistical sensitivity of the analysis (Lines 132–133).

It is important to specify whether the germination boxes were incubated under controlled temperature and photoperiod conditions, as well as to indicate the values used (Lines 134–

138).

In the field experiments, three replications per treatment were used, which corresponds to the minimum commonly accepted in agronomic studies. However, this level of replication may reduce the statistical power to detect differences among treatments, particularly when evaluating variables influenced by high environmental variability (Line 150).

Soil moisture monitoring or direct indicators of water stress during crop development are not reported, which represents an important limitation considering that the study evaluates responses potentially related to stress conditions. Including or discussing these environmental variables would help interpret the field results more accurately (Lines 150–173).

This section describes important edaphic differences between the experimental sites, which is a relevant aspect of the study design. However, it is not entirely clear how the “site” effect was incorporated into the statistical model used for data analysis. Since soil conditions can significantly influence crop agronomic response and the effectiveness of bio-inoculants, it is necessary to clarify whether site was considered a fixed or random effect in the model, or whether separate analyses were conducted for each location (Lines 154–161).

It is recommended to complement the analyses presented with the reporting of effect sizes in addition to statistical significance values. Including this type of metric would allow a clearer evaluation of the real magnitude of the differences among treatments and strengthen the quantitative interpretation of the results (Lines 180–184).

The manuscript could benefit from a more detailed analysis of site × treatment interactions, considering that the experiments were conducted in locations with contrasting edaphic conditions. In this regard, it would also be advisable to consider the use of mixed statistical models including site as a random effect, which would better capture environmental variability and improve the robustness of the statistical inferences (Lines 185–189).

Figure 5 presents a principal component analysis (PCA) integrating agronomic variables and evaluated treatments. However, several of the included variables do not show statistically significant differences among treatments, which may limit the biological interpretation of the analysis. In addition, the interpretation of the PCA in the text is relatively brief and mainly descriptive. It is recommended to deepen the discussion of variable loadings, the variance explained by each component, and the agronomic meaning of the observed groupings (Lines 246–253).

Although the discussion relates the results to some previous studies, in several sections it remains largely descriptive, mainly reiterating the results presented in the previous section. This section would benefit from the incorporation of mechanistic explanations that allow a deeper interpretation of the possible interactions among bio-inoculants, environmental conditions, and the physiological or agronomic response of the crop (Lines 278–320).

One limitation identified is the absence of a control treatment without inoculation and without fertilization (T0), which limits the causal interpretation of the observed effects. The inclusion of such a control is important to establish a baseline that would allow a clearer comparison of the real impact of the bio-inoculants and fertilization levels evaluated. It is suggested that this limitation be discussed in the manuscript or that the reason for its absence in the experimental design be justified (Line 424).

Microbial colonization or persistence of the inoculants in the root system was not evaluated, which makes it difficult to establish a direct relationship between the observed agronomic effects and the biological processes associated with the applied microorganisms (Line 425). Although this limitation is acknowledged in the discussion, it would be necessary to provide a clearer justification in the methodology explaining why it was not considered.

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